
Predicting Choice Among Car Safety Bundles

Final Report · Team 3 R-izzlers

1.185

public leaderboard, 3rd place

1.185

private leaderboard, 4th place

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Summary

Our graded submission, `cand_pool15050_final100.csv`, scored 1.185 on the public leaderboard (3rd place) and 1.185 on the private one (4th place), against a benchmark of 1.386. It is an equally weighted log-space pool of two independently built models, both calibrated to a single constant that we measured from the leaderboard itself rather than fitted on our folds. The public score and the private score were identical to three decimal places, consistent with a pipeline that did not overfit the public board, and every forecast we committed to version control before uploading came true. This report describes the validation machinery that every decision passed through (Section 2), the model (Section 3), the experiments that failed and what each failure taught us (Section 4), what the fitted models say about car buyers (Section 5), and how local scores related to the two leaderboards (Section 6).

1 Introduction

Each of 1,135 survey respondents faced 19 choice tasks. A task shows three configured safety bundles and a fourth option, “none of these”, and the respondent picks one. We predict the four choice probabilities for every task of 263 *new* respondents (4,997 tasks), scored by mean multiclass log loss. About 70% of the test rows form the public leaderboard and the remaining 30%, roughly 1,499 rows, form the private one that decides grades.

Three facts about the data shaped everything we built. First, the outside option is the single most chosen alternative: 30.2% of training choices, against 22.0, 25.0 and 22.7% for the three real bundles. Declining to buy is the modal behaviour, not an afterthought. Second, the 19 product attributes are ordinal tiers (three to seven levels; price has twelve), so treating levels as numbers imposes a linearity the data rejects (Section 5). Third, the test respondents are a different population: markedly wealthier (median income \$60k in training against \$80k in test; mean \$75k against \$151k), better educated, and far more concentrated in luxury vehicle segments (69% of test tasks against 9% of training tasks). The competition is therefore as much about surviving a population shift as about fitting choices.

2 Validation before modelling

Grouped folds. All cross-validation uses one fixed five-fold split grouped by respondent. Each person answers 19 tasks; a row-wise split would let a model learn “this person is frugal” from fifteen of their answers and be graded on the other four. Test respondents are strangers, so the local protocol must make every scored respondent a stranger too. A model’s score on these held-out people is its out-of-fold (OOF) log loss, the held-out score quoted throughout this report. Blend weights are refit five times, each time excluding the fold being scored, so no number we acted on had seen its own tuning data; we call a score produced under that discipline *nested*.

Noise floors before margins. Early in the project we measured how much scores move when nothing changes but the random seed, because a claimed improvement smaller than that movement is not a result. The three floors differ by an order of magnitude, and using the wrong one cost us once (Section 4):

comparison	correct yardstick	measured
two single models	seed sd of one model	0.00283
two blends	seed sd of the blend	0.00048
one model across folds	fold-to-fold sd	0.013

Model comparisons use paired tests with respondent-clustered standard errors: a genuine 0.005 gain is invisible in headline scores (fold sd 0.013) but resolves clearly when the two models are differenced on identical rows.

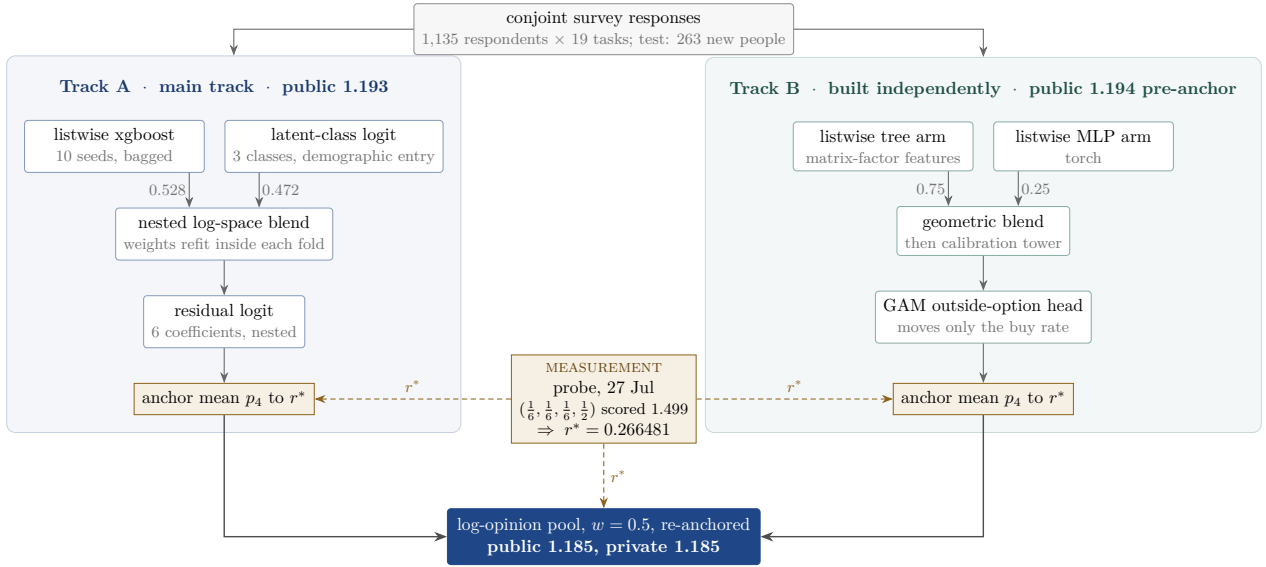


Figure 1: The graded submission, data to file. Rounded boxes are fitted models; square amber boxes are the one measured constant and the places it is applied; numbers on edges are blend weights. The pool step fits nothing: $w = 0.5$ was fixed before upload and never tuned. Track A as drawn scored public 1.193; track B’s public 1.194 predates its anchor step, which was added at pooling (Section 3.3).

3 The submitted model

The graded file, `cand_pool15050_final100.csv`, is a log-opinion pool (the two tracks’ log probabilities averaged, then renormalised) at fixed weight $w = 0.5$ of two tracks built independently by different team members, with no shared code and different fold constructors. Figure 1 shows the whole pipeline.

3.1 Track A: a two-member blend

Track A rests on three modelling choices, each verified by a paired test.

Part-worth coding (+0.020, $z = 11.8$, our largest single gain). Every attribute level receives its own utility rather than a constant step per level. The fitted price response drops steeply at first, then flattens and saturates at the top (Figure 2), a shape linear coding cannot express.

A listwise training objective (+0.006, $z = 4.3$). The competition metric is a softmax over the four alternatives within a task, so only relative utilities matter. We train the boosted trees with a custom objective whose gradient is $p - y$ taken within each four-row choice set, making the training loss identical to the scored loss rather than a pointwise proxy.

Discrete heterogeneity (+0.013, $z = 3.8$). A three-class latent-class conditional logit whose class membership is predicted from demographics only. Why this form of heterogeneity works when richer forms fail is the subject of Section 5.

The two members, a ten-seed bag of the listwise xgboost (out-of-fold 1.13682) and the latent-class logit with task-position terms (1.13863), are pooled in log space with weights 0.528 and 0.472 refit inside each fold, giving a nested score of 1.12819. We report these figures with a caveat we discovered ourselves and describe in Section 4: an encoding defect inflates them by roughly 0.008, and the honest nested figure is close to 1.136. Two further nested steps finish the track. The first is a residual logit, a small correction layer that nudges the blend’s log probabilities using six fitted coefficients on three features (relative price, an outside-option constant, and a bundle’s total feature count minus its best rival’s), each also interacted with a luxury-segment indicator. The second is the probe anchor described next. The residual logit was the sole survivor of roughly 160 candidate adjustments tested on 30 July: +0.00300 on the production folds and +0.00628 on an independent fold structure, the same sign twice but only about 1.7σ combined. We shipped it as the best-supported remaining gain, not as a proven one.

3.2 Track B: a tree and a neural network

The second track was built independently by a different team member and shares no code with track A. A listwise xgboost arm augmented with matrix-factorisation taste features (a rank-3 taste subspace fitted on training choices, then predicted from demographics so that unseen respondents receive taste estimates cold) is blended geometrically with a listwise multilayer perceptron. A calibration tower follows: per-segment shifts of the outside-option probability, a temperature that flattens overconfident predictions for luxury-segment respondents, a ridge residual logit, and finally a low-degree-of-freedom binomial GAM (generalised additive model) that supplies an independent probability of choosing “none”, mixed into the margin at 25%. The GAM head only moves how often someone buys; which bundle they would buy is carried through unchanged. Because this track’s fold construction differs from track A’s (Section 3.4), it has no out-of-fold score comparable to track A’s numbers. Its one measured public score is 1.194, before anchoring, and its anchored variant carried a pre-registered forecast of 1.185, band 1.183 to 1.191, which the pool superseded before it could be tested. Its blended arms score 1.153 out of fold on the track’s own partition (recomputed from the submitted artifacts; not comparable to track A’s differently partitioned numbers).

3.3 The measured constant r^*

On 27 July we spent one of our daily submissions on a measurement instead of an attempt, applying a known leaderboard-probing idea to our setting. The file `probe_alt4.csv` predicts the same vector $(\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{2})$ on every row. For a constant prediction the log loss is exact algebra in one unknown, the fraction r of scored rows whose true answer is “none”:

$$L = (1 - r) \log 6 + r \log 2 = \log 6 - r \log 3 \quad \implies \quad r^* = \frac{\log 6 - 1.499}{\log 3} = 0.266481.$$

The board displays three decimals, so the returned 1.499 pins the public none-share to a band of roughly 0.2660 to 0.2669; we anchored to the displayed value’s point solution. **This number is measured on the test set, not fitted on our folds**, so we expected it to transfer more reliably than any fold-fitted correction; the private result was consistent with that expectation, though it cannot isolate the anchor’s contribution. Both tracks were tilted so their mean predicted probability of “none” (p_4) equals r^* , using a single global odds multiplier that leaves every within-purchase conditional probability unchanged. Track A moved little (+0.00104 expected). Track B, we found during a final audit, had shipped with mean $p_4 = 0.211$ and had never been anchored; correcting it was worth roughly 0.0097 and accounts for most of the improvement from 1.193 to 1.185.

3.4 The pool, and why w was never tuned

The two tracks use different fold constructions (adjusted Rand index 0.002, effectively unrelated partitions), so no clean nested estimate for tuning w was available from the two prediction sets we had; building one would have meant refitting both tracks under a shared fold structure, which time did not permit. We fixed $w = 0.5$, which spends no selection budget, and relied on a bound rather than a forecast: writing the pooled loss as $(1 - w)L_A + wL_B + \mathbb{E}[\log Z_w]$ with $Z_w = \sum_{k=1}^4 A_k^{1-w} B_k^w$ taken over the four alternatives of each row, Hölder’s inequality gives $Z_w \leq 1$, hence

$$\text{loss}(\text{pool}) \leq (1 - w)L_A + wL_B \leq \max(L_A, L_B)$$

on every set of rows, including the private 1,499. Pooling only pays if the parents genuinely differ, so we measured that too: the tracks’ within-purchase predictions sit 0.0129 apart on average, 2.3 times the 0.0055 floor obtained by reseeding a single model, and they disagree on 96.6% of rows. Two distinct models, not one model twice. When exactly one submission counts and public rank says little about private rank (Section 6), a bounded worst case was worth more to us than a marginally better point forecast.

3.5 The field the members beat

Before track A’s two blend members were chosen, nine model families were fitted to the same data under the identical grouped validation of Section 2. The blend members are simply the last two

standing. Figures are plain out-of-fold log loss; rows involving trees inherit the inflation caveat of Section 4.

model family	OOF	verdict
listwise xgboost, 10-seed bag	1.13682	blend member 1
latent-class logit, 3 classes	1.13863	blend member 2
nested logit, {bundles} vs {none}	1.15681	indistinguishable from plain logit
part-worth conditional logit	1.15686	one-class special case of member 2
elastic-net logit, demographic interactions	1.16718	shrinkage cannot substitute for part-worths
two-stage: buy at all, then which bundle	1.17169	splits one utility scale in two
mixed logit, continuous coefficients	1.17281	predicts a population average (Section 5)
wide four-class xgboost	1.17456	cannot compare within a choice set
hierarchical Bayes	1.23703	cold start (Section 5)
nested blend of the two blend members	1.12819	honest $\approx$ 1.136 (Section 4)

Read top to bottom, the last column is the course in miniature: what separates the winners is not capacity but fit to the structure of the decision, a shared utility scale across the four alternatives, ordinal attribute tiers left free to bend, and taste heterogeneity routed through something observable about a stranger.

4 What failed, and what it taught us

More of what this report knows came from experiments that failed than from the three ideas that worked. Three episodes, in increasing order of cost, and then what they changed.

4.1 The seed is a hyperparameter nobody tunes

With subsampling at 0.8 and a random early-stopping holdout, refitting the identical xgboost configuration under ten seeds moved the out-of-fold score across a range of 0.0090 (sd 0.00283). That sd is larger than most margins in our experiment log. It invalidated one accepted result: a monotone price constraint (forcing predicted utility never to rise with price), adopted for an apparent +0.00172, retested paired across ten seeds at -0.00034 with 95% CI $[-0.00159, +0.00092]$, winning five seeds of ten. The constraint did nothing; we had been carrying the same model twice under different names. Averaging over seeds is also the one improvement that cannot overfit, since it selects nothing; yet at the blend level it was worth only +0.00029, because a blend is itself a variance-reduction device and its members already absorb most of each other's seed noise. **Bagging and blending are substitutes, not complements.**

4.2 A leak that honoured the fold rule

One experiment scored 1.0996 when nothing else we had built had broken 1.128: an apparent leap of 0.03, half again our largest genuine gain, from a single derived feature: the average residual (observed minus predicted) of the other respondents who saw the same choice-set design, always computed from other folds. The fold rule was honoured; the leak was in the baseline. Each reference respondent carried a prediction from a model trained on folds that included the scored fold, and a deep tree can partially memorise a design, so subtracting the baseline handed the target its own label back with the sign flipped. Rebuilt with the leak made structurally impossible (for every ordered pair of folds, fit excluding both), the gain fell to -0.00430 ($z = -4.54$). The clean version had passed both of our leak detectors, and passed them more convincingly than the leaky one. A signal test can establish that a feature contains information; it cannot establish that a flexible learner's gain comes from that information rather than from a leak riding alongside it. Only a construction that makes the leak impossible settles the question.

4.3 A structural leak that survived 47 iterations

A design-share encoding (per-design empirical choice shares, built with the standard leave-one-fold-out discipline) sat in the production model for 47 iterations and appeared to be worth +0.022. It was a leak of a kind our checks were not designed to catch: the encoding function was applied

once, before the cross-validation loop, so training rows were encoded from a reference set that contained the scored fold. Because the defect is baked into the construction, it appears *uniformly* in every fold; per-fold consistency, our standard screen, proves nothing about honesty. Three tests do work, and we used all of them: build an honest twin with matched support (honest value -0.00596 , $z = -3.53$); check whether the gain grows with model capacity, since real effects saturate while leaks do not (from -0.0004 at depth 4 to $+0.078$ at depth 10); and ablate the feature entirely. The retraction also voided our hyperparameter sweep, whose honest ranking reversed, and explains a period in which local gains produced public *regressions*: two submissions built on contaminated validation improved locally by 0.005 and 0.016 and scored 1.209 and 1.205 publicly, against 1.199 before them. Five pre-registered gates had passed for one of those submissions; every gate was computed on reweightings of the same contaminated out-of-fold predictions. **Many tests sharing one assumption are one test.**

4.4 What the episode changed

After the discovery, plain cross-validation scores were treated as an anti-signal below 1.128 (past that point, a better local score predicted a worse public one), and decisions moved to a segment-reweighted out-of-fold metric (training respondents reweighted to the test segment mix) that tracked the public board to 0.001 when leak-free. Its effective sample is only 208 of 1,135 respondents, so we used it to judge and never to fit; fitting on a reweighted objective was refuted three separate times in our log, each time because the effective-sample collapse costs more variance than the bias it removes.

5 What the data says about buyers

5.1 Three kinds of buyer, and what the aggregate curve hides

The latent-class model splits respondents into three near-equal segments (shares rounded to one decimal), with membership predicted from demographics; the strongest drivers are annual mileage, night driving, education and parking situation:

segment	share	price response	outside-option constant
value-conscious buyers	36.6%	-1.65 , steady decline	-1.40 (buys readily)
price-indifferent buyers	32.1%	$+0.50$, flat to positive	-1.30 (buys readily)
reluctant non-buyers	31.4%	-0.60 , mild	$+0.17$ (inclined to decline)

This reframes Figure 2. A single logit fitted to everyone shows the flattening curve, which invites a psychophysical reading (people perceive price ratios, not differences). The segment model suggests much of that flattening is a mixing artefact: average a steep response, a flat one and a mild one, and the aggregate bends even though no segment's own curve does. Whichever reading is right, the pricing implication stands: at the top of the range one price step is nearly invisible to buyers, so shaving a single tier off a premium bundle buys almost nothing. One caveat needs stating: a positive price coefficient for the second segment is not behaviourally standard, and we cannot distinguish price acting as a quality signal from residual confounding with bundle richness, so we present it as

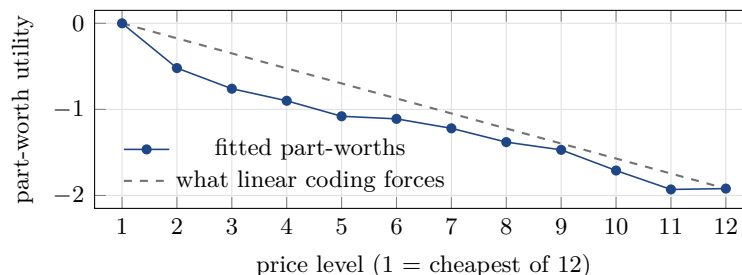


Figure 2: Price response flattens: the first step up from the cheapest level costs 0.52 utility, later steps average 0.13, and the top two levels are indistinguishable. Freeing this shape was worth $+0.020$ log loss, the largest single gain of the project.

an estimated pattern rather than established psychology. Commercially the third segment is the finding: **nearly a third of respondents belong to a segment inclined to decline at every price we tested**, so discounting is unlikely to reach them, and the leverage sits in the first two segments.

5.2 Heterogeneity is only usable in discrete form

Mixed logit with continuous random coefficients estimates large heterogeneity (price sensitivity varying six-fold across the interquartile range, outside-option constant sd 2.47) yet scored 1.173 against 1.157 for the fixed-coefficient model and earned zero blend weight. Hierarchical Bayes, the textbook tool for panel conjoint, made the reason quantitative. Scoring identical rows two ways:

prediction for the same rows	log loss
each respondent's own fitted tastes (posterior mean)	0.352
population-averaged tastes, what a new respondent gets	1.229

The 0.877 gap is the cold-start problem stated as a number: everything HB learns about individuals is structurally unavailable for the 263 strangers it is graded on. Its full overall score, 1.23703 in the table of Section 3.5, makes it our worst model. The latent-class model wins with the same idea because it routes heterogeneity through a bottleneck: demographics predict a three-way membership ($(3 - 1) \times 29$ demographics = 58 parameters) rather than shifting all 73 part-worths per demographic ($6 \times 73 = 438$), so HB reached eight times the parameters from a fifth of the demographics. Ablating demographics *improved* HB from 1.237 to 1.164 while the same information carries 93% of the latent-class model's gain. **Heterogeneity helps only to the extent it can be recovered from what you observe about a stranger**, and a low-dimensional discrete route is what makes that recovery estimable from 1,135 people.

5.3 Opting out is not a separate decision

Since “none” is the modal choice, we tried treating it specially twice: a two-stage model (buy at all, then which bundle) scored 1.172 ($z = -11$), and a nested logit isolating the outside option matched plain multinomial logit to three decimals, so the textbook correction for independence of irrelevant alternatives was not the binding constraint either. Declining is governed by the same utility scale as choosing between bundles; what varies across people is the threshold, which is what the latent-class constants capture.

5.4 Fatigue arrives as sharpened price sensitivity

The walk-away rate rises steadily across the 19 tasks of a survey session (Figure 3), and the data favour one explanation over its rival. Adding task-position terms to the latent-class utility was worth +0.0053 ($z = 6.3$) and replicated on an independent fold structure (+0.0042, $z = 4.7$). A specification allowing only the price coefficient to steepen with task position reproduces 105% of the observed drift in declining, while a pure scale-drift account (respondents answering more noisily) reproduces essentially none of it. Late in a session people are not choosing randomly; they are

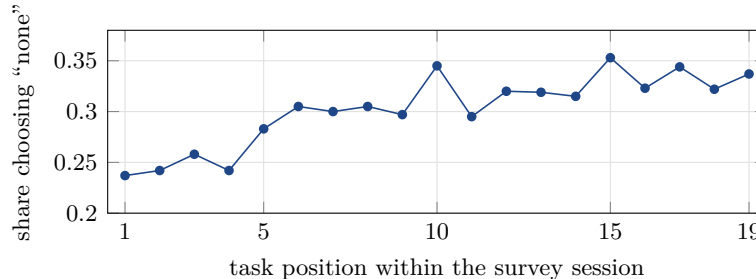


Figure 3: Raw training data, no model: the share of tasks answered “none” climbs from 23.7% on a session’s first task to above 33% late in the session, roughly ten percentage points of drift. Section 5.4 weighs two explanations for it.

behaving like sharper price hagglers, and since the outside option is always the cheapest thing on offer, sharper price response mechanically raises refusals. The effect retains 119% of its value under reweighting to the test population, the strongest retention of any gain we measured.

5.5 The population shift, and a prediction it let us test

Because test respondents are wealthier, we added explicit price-by-income interactions (linear models extrapolate; trees cannot) and audited every accepted gain by reweighting training respondents to the test income distribution: part-worth coding retains 101% of its value, the listwise objective 112%, the latent-class model 79%. The membership channel also made a falsifiable claim: routing wealthier respondents toward buying segments, the latent-class model predicted a test walk-away rate of 0.223, far below the training rate of 0.302. The probe measured 0.266. The direction was right, wealthier people do decline less, and the magnitude was overshot, which is why we anchored to the measured value rather than to any model's opinion.

5.6 Where the remaining headroom lies

We can state where the remaining accuracy sits and why we stopped searching for it. An oracle handed each respondent's own observed choice rates scores 0.986 against 1.130 for the blend as it stood when we measured this, so under this construction the remaining loss lives in a 0.144 gap of person-specific behaviour. Decomposing choice variance, only the outside option is strongly personal (33% between-respondent, against 5 to 7% for the bundles), and demographics as we fitted them explain just 11.2% of that between-respondent variation. **Roughly 89% of the between-respondent variation in walking away is unexplained by everything we observed**, and no model fitted on these features can recover variation they do not carry. Choice-set identity alone also leaves most of the error in place: a predictor built purely from design identity scores 1.307. We therefore spent the final week on measurement and variance control rather than new models; Section 6 shows that is where the remaining score was.

6 Public versus private

6.1 Local scores predict the board poorly, and we measured how poorly

local nested OOF	public	offset	transfer of the local gain
1.17683	1.223	+0.046	(first submission)
1.13878	1.201	+0.062	~58%
1.13556	1.201	+0.065	invisible (rounding floor)
1.13044	1.199	+0.069	~24 to 39%

Cross-validated scores were optimistic by a stable 0.05 to 0.07: partly the documented population shift, partly ordinary selection optimism from testing many ideas against one fold structure. Transfer of local gains decayed from 58% to about a third, then went negative during the leak era of Section 4 (two local improvements returned 1.209 and 1.205). The public board also has a resolution floor: a 0.0032 local improvement produced no visible public change at three displayed decimals. Our response was the segment-reweighted metric (which tracked the board to 0.001 when leak-free) for judging, and leaderboard measurement rather than leaderboard climbing for calibrating.

6.2 Three forecasts, committed before upload

Both forecasts below were committed to version control before their uploads, so neither could be revised after the fact. A third pre-registration, r^* with its derivation, faced its test only at the close, on private rows it had never touched.

quantity	forecast	band	returned
track A final file	1.1930	point forecast	1.193
graded pool	1.186	1.184 to 1.189	1.185

6.3 What the close settled

The private score equalled the public score to three decimal places, 1.185, although the private subset's none-choice share alone carries a sampling sd of about 0.011 on roughly 1,499 tasks. Every constant calibrated on public rows, above all r^* , then faced rows it had never touched. One rounded aggregate cannot isolate any single constant's transfer, but the result is what intact transfer predicts: no detectable deterioration at the board's displayed precision.

The rank still slipped from 3rd to 4th on an unchanged score, and that was predictable too, because ranking noise and score noise are different quantities. The sd of one team's absolute score on 1,500 rows is about ± 0.02 , which tempts the conclusion that nothing smaller can matter; but all teams are scored on the same rows, so rank differences are paired, the shared row-difficulty shocks cancel, and the relevant standard error is 0.006 to 0.012, driven only by where teams disagree. Our lead over the nearest rival was about one to two paired standard errors: a probable win, never a safe one. **The rank moved; the model did not.** We regard the zero drift, the forecasts landing, and the one-place rank slip as a single consistent picture of a pipeline that was calibrated, not lucky.

7 Limitations

The irreducible noise is large: per-respondent mean loss ranges from 0.59 to 2.71, and most of what drives the outside option is unobserved (Section 5). Two of our gains are population-specific and would not survive a new experimental design or population: anything routed through choice-set identity, and the latent-class membership channel (79% retention under income reweighting). The three-segment structure is one of several that fit comparably; the existence of heterogeneity is robust, the specific boundaries are not. The shipped tree still contains the defective encoding step of Section 4: we measured its removal as board-neutral and judged late rebuilding an unnecessary risk, but a new project should not copy that feature step. Finally, we optimise average log loss only; no subgroup is separately calibrated, and predictions are least reliable where support is thinnest: the wealthy luxury segments the test set concentrates on.

8 Reproducibility

The code package carries two different guarantees. Fresh training reconstructs both modelling procedures approximately: track A via `model/run_all.R` followed by the residual logit and anchor, track B via its own `run_all.R`. Multithreaded xgboost and torch make bit-exact retraining impossible across machines, so retrained tracks land within about 10^{-3} of the scored ones. Separately, a deterministic reconstruction of the graded submission, built by `build_candidates.R` from the two pinned track outputs and the probe algebra, agrees with the scored file to within 2×10^{-5} . Two traps cost us real time and are documented in the package: xgboost 3.x returns a matrix from `predict` where 2.x returned a vector, silently scrambling reshaped predictions; and in a conditional logit anything constant within a choice set is unidentified, so the price reference level must sit on a real bundle, with a rank-revealing QR on the task-demeaned design catching the rest.

9 Conclusion

Three first-week structural choices, part-worth coding, a listwise objective and demographically routed discrete heterogeneity, produced nearly all the accuracy this project earned. The weeks that followed added more knowledge than score: an account of where the remaining accuracy sits, two dissected leaks, and a demonstration that a leaderboard can serve as a measuring instrument rather than a slot machine. The submission carried a worst-case bound instead of a bet and landed where its pre-registered forecast said it would, on both halves of the test set. For future cohorts, one sentence: measure the noise floor before trusting a gain, build the honest twin of every clever construction, and when the test set will be strangers, validate as if every scored respondent were one.